Customer Personality Segmentation

Make Sense of Unstructured Data

3/19/26 ~ Sydnee Sampson-Blackwell

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Business Problem Overview

Business Context: Understanding customer behavior is pivotal for enhancing satisfaction and revenue. A leading retail company seeks deeper insights into customer profiles to unlock personalization opportunities and sustain a competitive edge in an intensifying market.

The Challenge: Moving away from generic strategies to targeted approaches is essential for fostering long-term relationships and improving campaign effectiveness.

Objective: Identify distinct customer segments to develop personalized marketing campaigns and increase conversion rates. Create effective retention strategies for high-value groups and optimize resource allocation such as inventory and pricing.

Responsibility: Analyze customer data using machine learning to provide actionable insights into the unique characteristics of each identified segment.

Data Overview

Dataset: 2240 records with attributes covering demographics, spending categories, and purchase channels.

Data Types: Mostly numerical integers and floats, with three object columns for Education, Marital Status, and Enrollment Date.

Average Income: The average household income is approximately $52,247.

Missing Values: Only the Income column had missing data, specifically 24 null values which were handled before analysis.

Duplicates: Checked for duplicate records and confirmed there are 0 identical rows in the dataset.

Outliers: Spending variables (Wines, Meat, Fish, Fruits, Sweets, Gold) and Income contain significant high-end outliers, consistent with a small segment of high-value customers identified in EDA.

Scaling: StandardScaler was applied to all features before clustering to ensure no single variable dominated the distance calculations due to differing units or ranges.

Data Overview - Statistical Summary

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts
count	2240.000000	2240.000000	2216.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000
mean	5592.159821	1968.805804	52247.251354	0.444196	0.506250	49.109375	303.935714	26.302232	166.950000	37.525446	27.062946
std	3246.662198	11.984069	25173.076661	0.538398	0.544538	28.962453	336.597393	39.773434	225.715373	54.628979	41.280498
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	2828.250000	1959.000000	35303.000000	0.000000	0.000000	24.000000	23.750000	1.000000	16.000000	3.000000	1.000000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	49.000000	173.500000	8.000000	67.000000	12.000000	8.000000
75%	8427.750000	1977.000000	68522.000000	1.000000	1.000000	74.000000	504.250000	33.000000	232.000000	50.000000	33.000000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.000000	259.000000	263.000000

Univariate Analysis

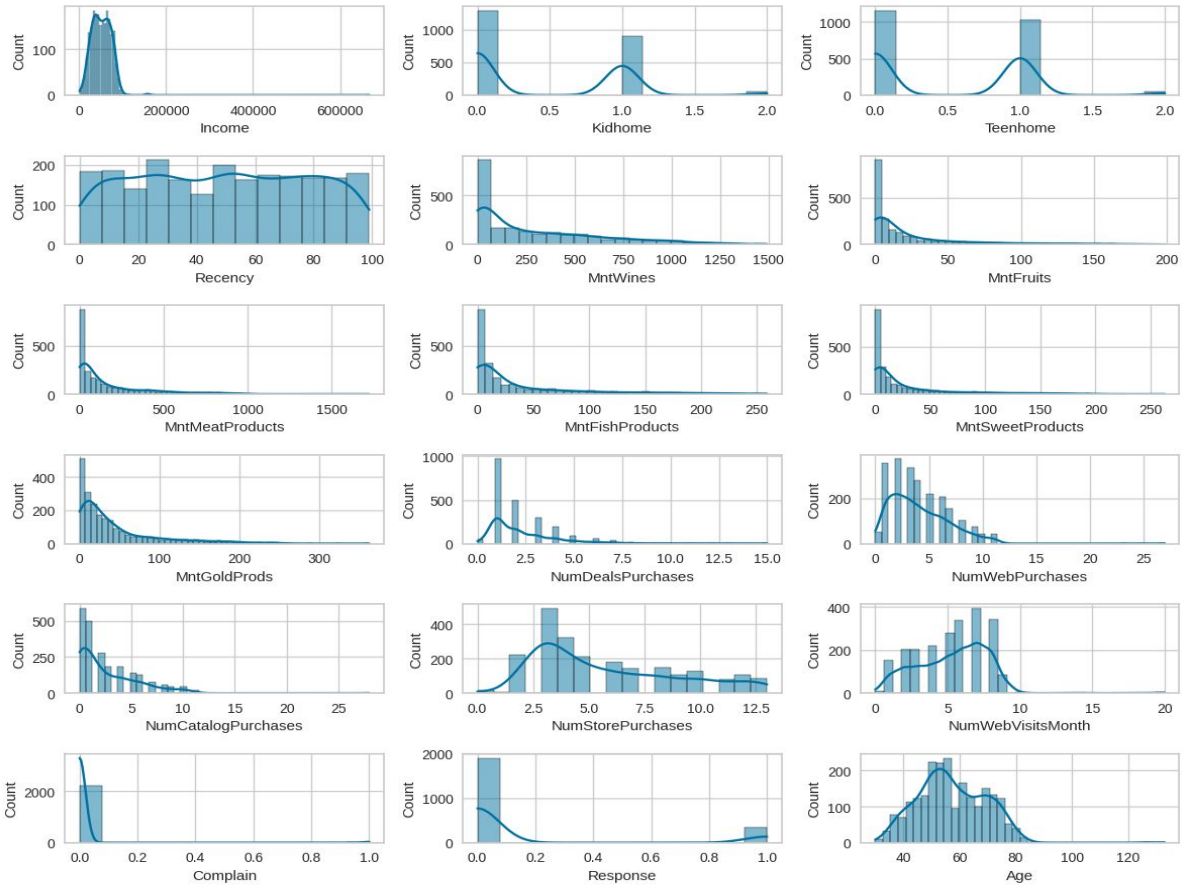
Demographics: Income and Age are roughly normally distributed, though Income has extreme high-end outliers.

Spending Categories: All spending variables like Wines, Fruits, Meat, Fish, Sweets, and Gold show strong right-skewed distributions, indicating a small number of high-value shoppers.

Purchase Channels: Store purchases are the primary channel for most customers, followed by Web purchases. Catalog purchases have the lowest frequency across the board.

Website Engagement: Most customers visit the website 5 to 7 times per month, regardless of their actual spending level.

Univariate Analysis - Histogram Of Each Column



Multivariate Analysis

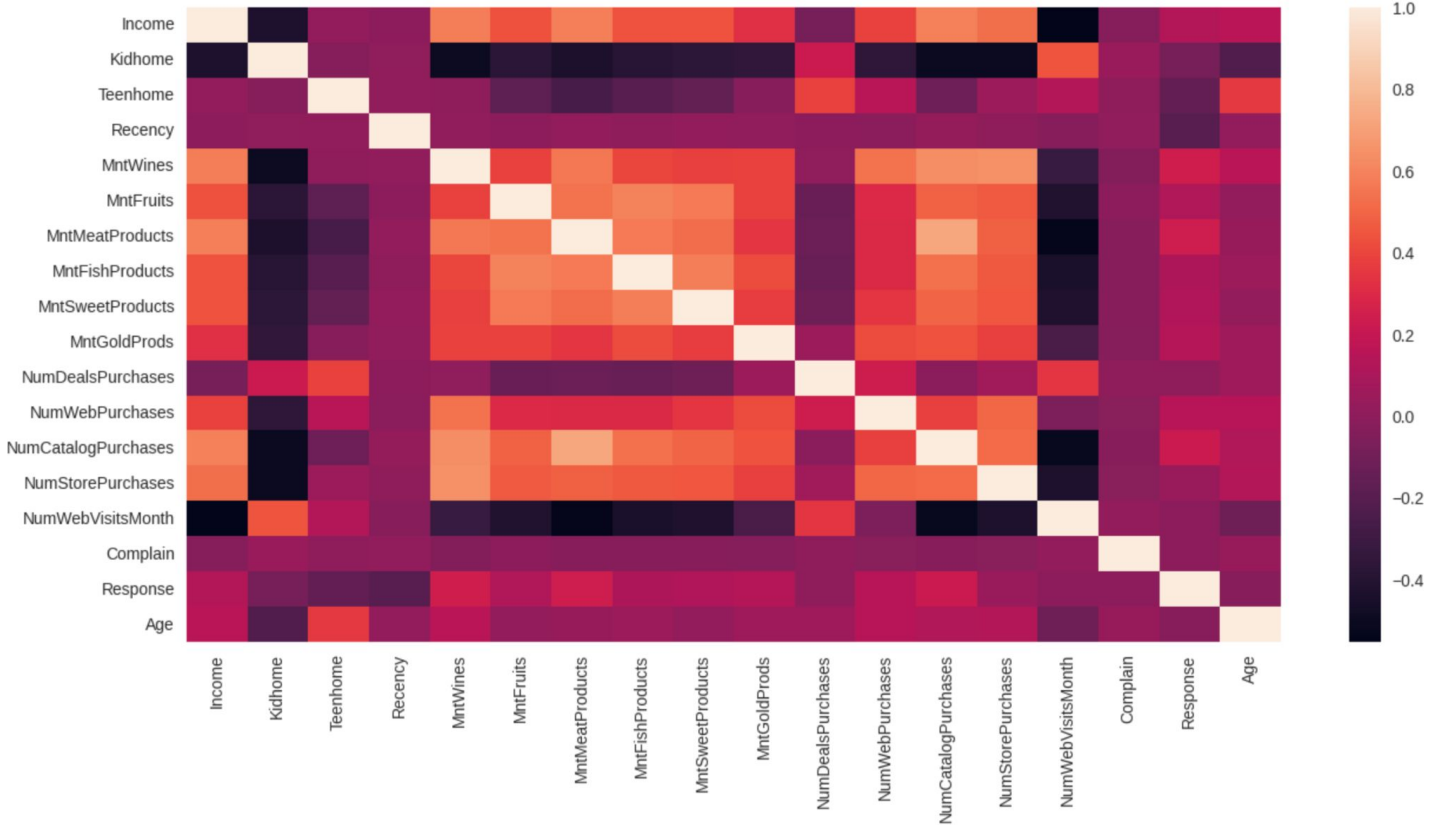
Income and Spending: Strongest positive correlation observed between Income and high-end categories like MntWines and MntMeatProducts.

Family Dynamics: Negative correlations found between the number of children/teenagers at home and spending on luxury items like Wine.

Purchase Channels: Positive correlation between Store, Web, and Catalog purchases suggests that high-value customers engage across all available channels.

Web Visits: Interestingly, web visits show a negative correlation with actual spending, suggesting lower-income customers browse more frequently without purchasing.

Multivariate Analysis Heat Map



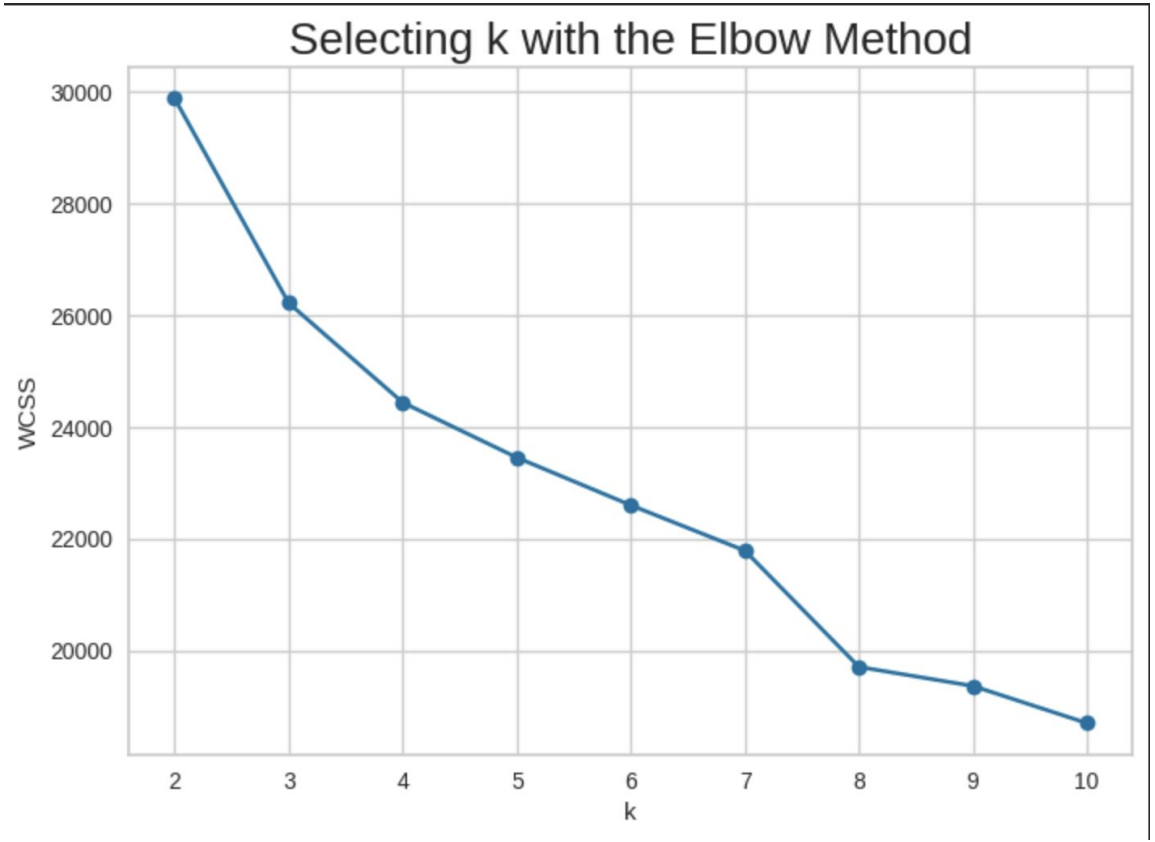
K Means Clustering:

Elbow Method: Analyzed the WCSS for k-values 2 through 10. A distinct elbow was observed at k=4, indicating an optimal point for cluster separation.

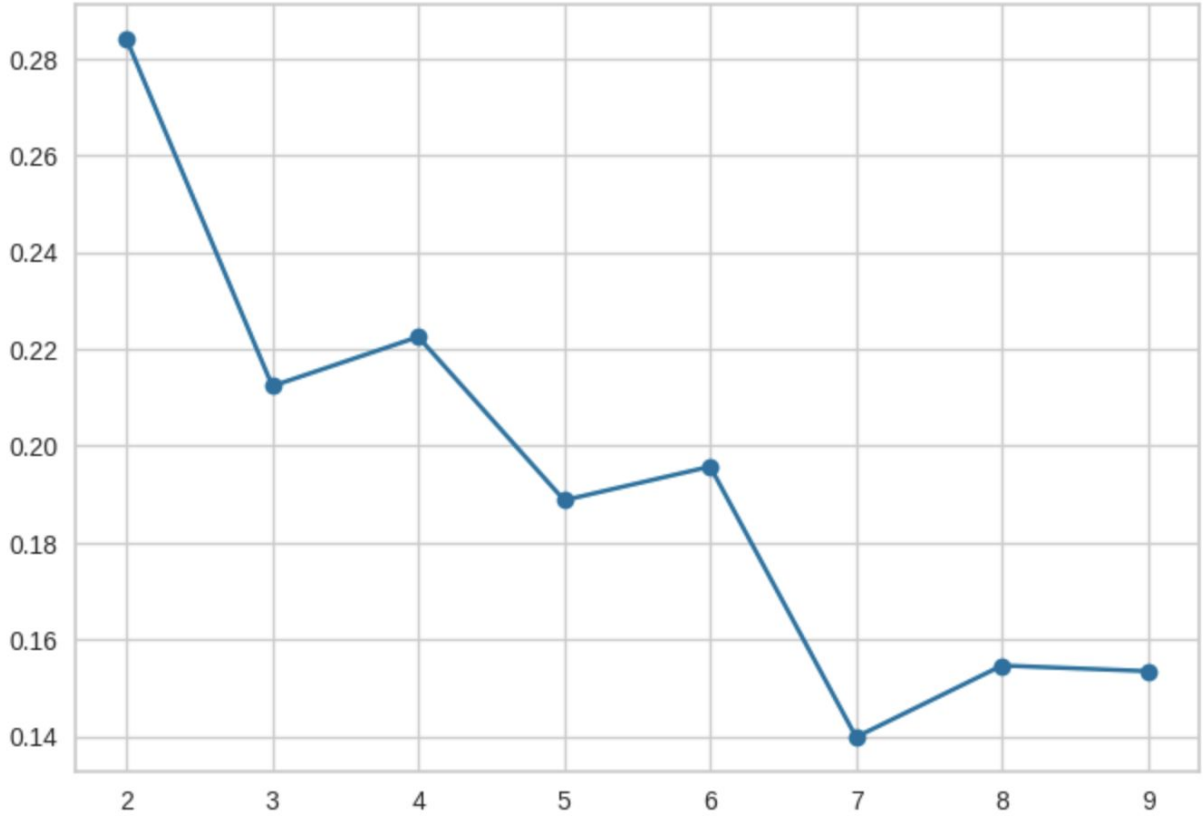
Silhouette Scores: Validated the number of clusters using silhouette scores. While k=2 had the highest score, k=4 provided a better balance between mathematical distinctness and actionable business segmentation.

The final model with 4 clusters was fitted to the scaled data with an execution time of approximately 9 milliseconds, demonstrating high computational efficiency.

Elbow Method:



Silhouette Scores:

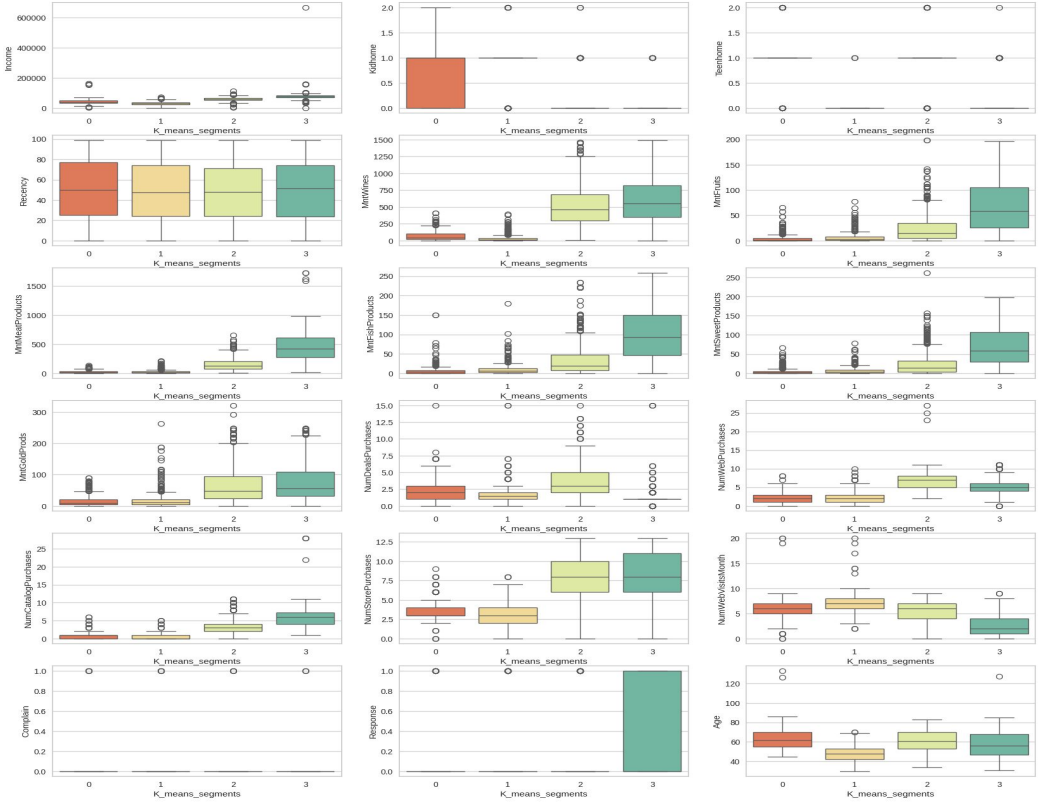


Cluster Profiling and analysis:

- **Cluster 1 High Spenders:** High income and heavy spending on Wine. Low deal sensitivity. Prefer Catalog and Store shopping.
- **Cluster 2 Budget Families:** Households with high numbers of children who rely heavily on discounts and deals. Lower luxury spending.
- **Cluster 0 Opportunity Segment:** Frequent web visitors with low conversion rates and lower income. High interest but low current spend.
- **Cluster 3 Balanced Shoppers:** Mid-range income with steady, balanced purchasing habits across all channels and categories.

Cluster Profiling: Boxplot of numerical variables

Boxplot of numerical variables for each cluster in Kmeans Clustering



Cluster Profiling: Barplot of all variables

Business Recommendations:

- **Retention:** Create a premium loyalty club for Cluster 1 to secure high-margin sales.
- **Promotions:** Launch family-sized bundle deals for Cluster 2 via email campaigns.
- **Conversion:** Use first-purchase discount codes to turn Cluster 0 web traffic into actual revenue.
- **Convenience:** Promote Buy Online Pick Up In Store services to retain the balanced shoppers in Cluster 3.

Great Learning

Happy Learning !